ENIGMA:

Breaking Germany's Secrets



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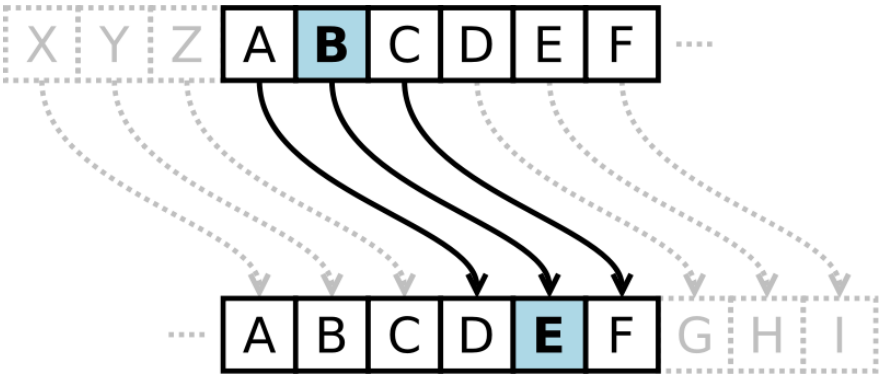
Professor Goulding

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Cryptography, the practice and study of hiding information, has been used extensively by secretive government agencies for centuries. A very well-known primitive encryption system was the Caesar cipher, used by Julius Caesar around approximately 45 B.C. ("History", 2007, par 6). This was a very low level of encryption, and only shifted each letter of the alphabet over a specified number of places.



Although cryptography and encryption have been in existence throughout history, it had not developed into anything particularly important until the 19th century. Edgar Allan Poe, an American poet, used systematic methods to solve ciphers in the 1840's ("History", 2007, par 13). Later on, he wrote an essay on methods of cryptography that provided useful insight for British code breaks to crack German codes during World War I.

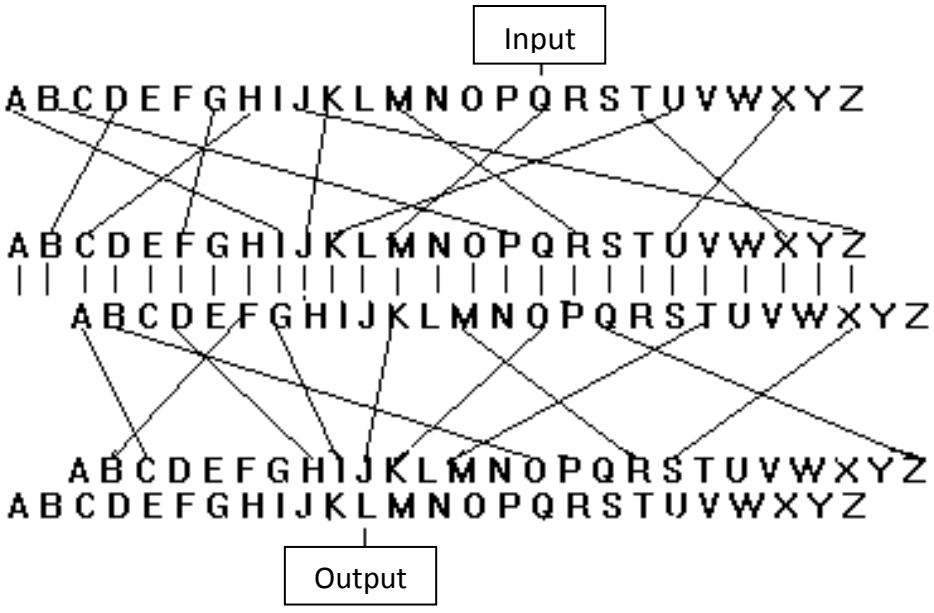
Gilbert Vernam, in 1917, proposed a teletype cipher, encrypting characters with a previously-prepared key. This led to development of the "one time pad", an algorithm where plain text is combined with a random key that forms a string used only once ("History", 2007, par 14).

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CIHJT UUHML FRUGC ZIBGD BQPNI PDNJG LPLLP YJYXM
DCXAC JSJUK BIOYT MWQPX DLIRC BEXYK VKIMB TYIPE
UOLYQ OKOXH PIJKY DRDBC GEFZG UACKD RARCD HBYRI
DZJYO YKAIE LIUYW DFOHU IOHZV SRNDD KPSSO JMPQT
MHQHL OHQQD SMHNP HHOHQ GXRPJ XBXIP LLZAA VCMOG
AWSSZ YMFNI ATMON IXPBY FOZLE CVYSJ XZGPU CTFQY
HOVHU OCJGU QMWQV OIGOR BFHIZ TYFDB VBRMN XNLZC
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German electrical engineer Arthur Scherbius patented the Enigma in 1918, in Berlin, Germany (Sale, n.d., "Principle", par 16). The Enigma is a cipher machine used to encrypt, as well as decrypt, messages containing sensitive data. Scherbius' ideology was that with each key press, the input character would be substituted into another. These substitutions, of which occurred multiple times, were achieved by electrical connections.



Arthur Scherbius

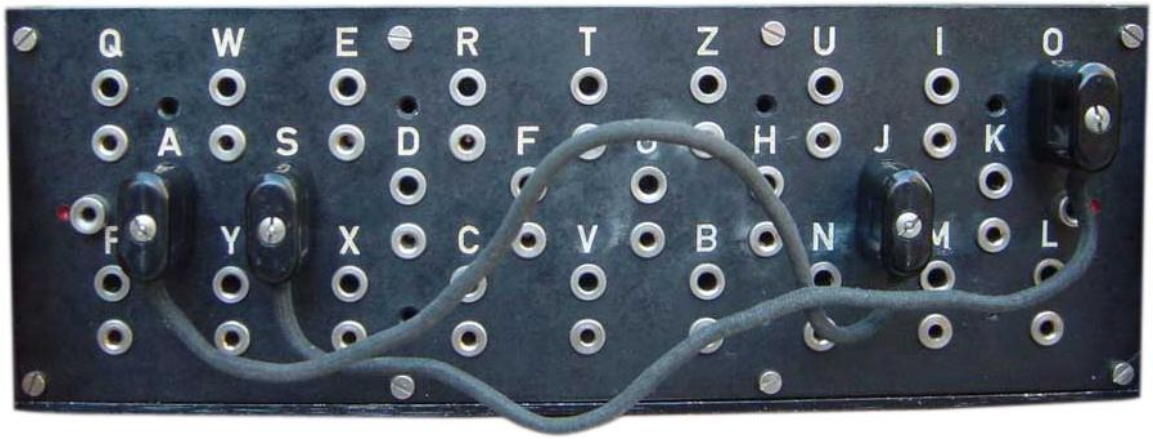


Scherbius began selling the Enigma commercially to banks and businesses. In 1924, the German armed forces adapted it for military use during World War II and believed it to be unbreakable (Ellis, 2007, pars 4-5).

The Enigma is more than just a direct substitution cipher. It has a combination of both mechanical and electrical systems. The input keyboard, a set of rotors, and a stepping mechanism for the rotors make up the mechanical portion (Enigma, 2007, pars 5-6). The keyboard, used only for input, was set up much like a typewriter, in that it had a fixed arrangement of raised keys. The arrangement very closely resembled a modern-day "QWERTY" keyboard, with the exception of the Y and Z being switched, and the P and L being moved to the bottom row, as shown by this diagram:

Q	**W**	**E**	**R**	**T**	**Z**	**U**	**I**	**O**
	A	**S**	**D**	**F**	**G**	**H**	**J**	**K**
P	**Y**	**X**	**C**	**V**	**B**	**N**	**M**	**L**

When a key is pressed, a path is created from the battery to the plug board. The plug board is similar to a telephone switch board. Characters that are to be switched must be plugged into one another in a one-to-one fashion. If one plugs A into X, then every A input at this point in the system is switched into an X, and every X into an A. Up to thirteen connections can be made, switching every letter in the alphabet. The plug board was used to switch characters directly after input, and directly before output.

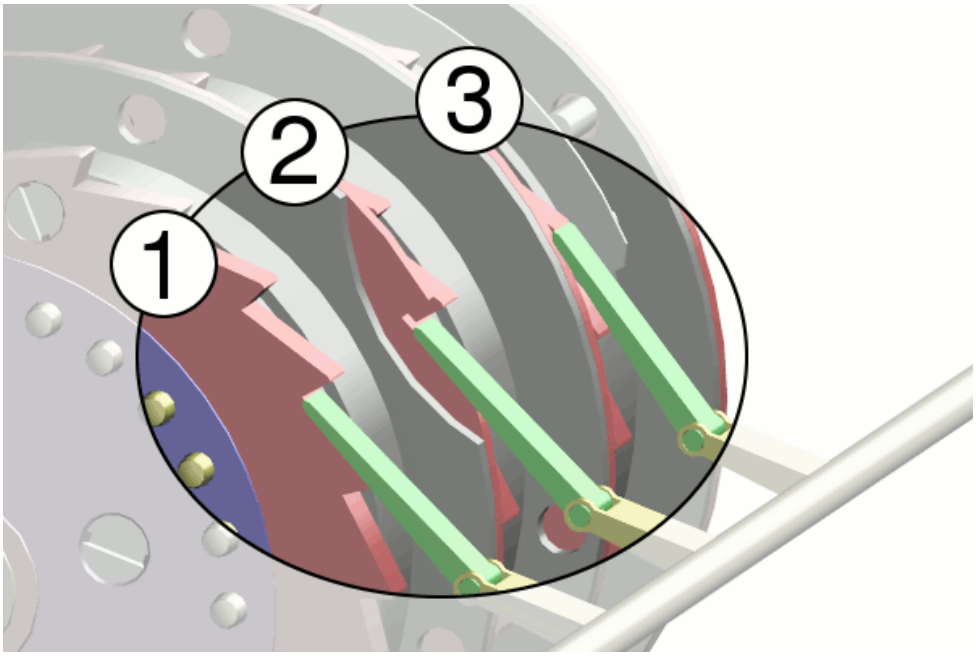


A plug board mapping A to J, S to O, and vice-versa.

The next portion of the Enigma system is the most important, and also the most complex: the rotors. Most versions of the Enigma had three slots to drop in mechanical wheels, which contained a notch so they would occasionally function as "gears". There were initially three rotors, labeled I, II, and III. Each of these rotors has its own specific internal wiring. When electricity passes through the rotor, it changes the signal into another letter.

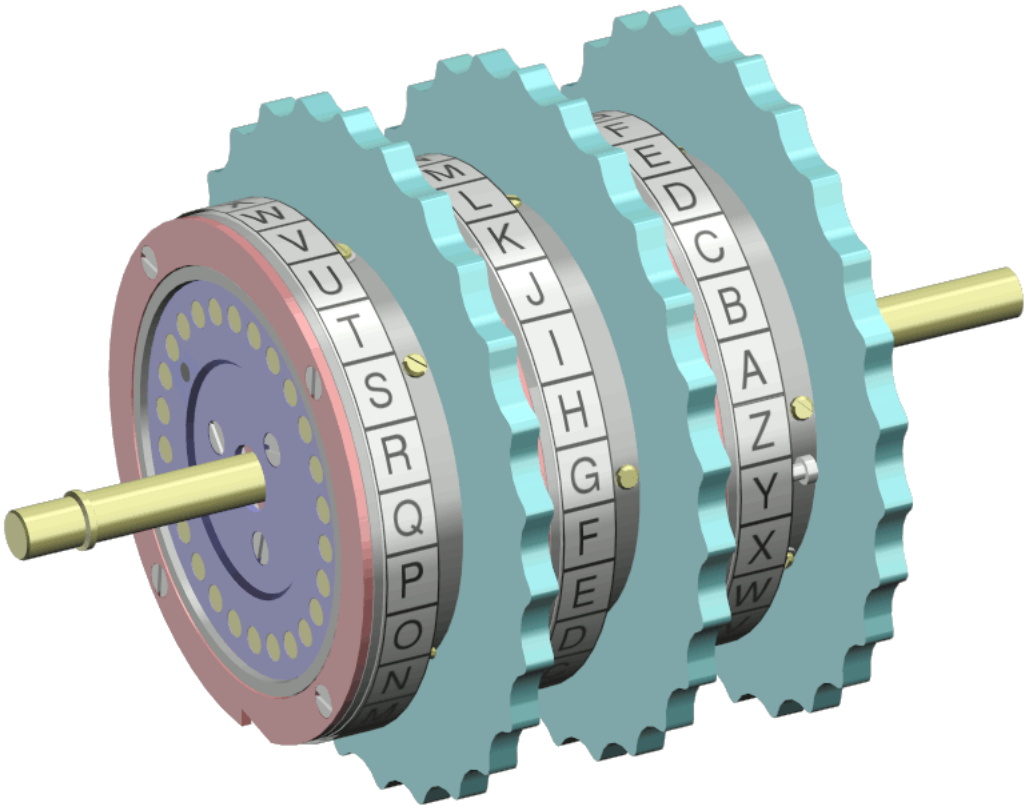
What makes this rotor system so complex, and provided such a large challenge to break, was that each rotor turned once every so often. This produced a polyalphabetic cipher, a substitution based on multiple instances of the alphabet ("Enigma", 2007, par 15). Every time a key was pressed, the right-most rotor stepped forward once. Once during each rotor's full rotation, the rotors clipped into a notch on the rotor to their left, and stepped it one place. Because of this dynamic changing of circuits, an input of "AAAAA" would produce up to five separate output characters.

The Enigma's rotors were implemented with a ratchet and pawl mechanism. Each rotor contained 26 teeth. There were also three pawls, pivoting bars that engaged the ratchet. With each key press, the three pawls simultaneously pushed, always turning the right-most rotor. In the occasion that the middle and left-most rotors' notches lined up with the pawls, they would also be pushed. No current was sent through the circuit until the rotors had been moved into place.



However, when the left-most rotor's notch was activated simultaneously by the middle, a phenomena called "double-stepping" occurred, resulting in the middle rotor being turned again. Because of this, the equation for the amount of encryption actually performed is 26×25×26, yielding 16,900 possibilities during the first pass through the rotors ("Enigma", 2007, pars 18-19).

Three rotors in sequence as they would be in the Enigma

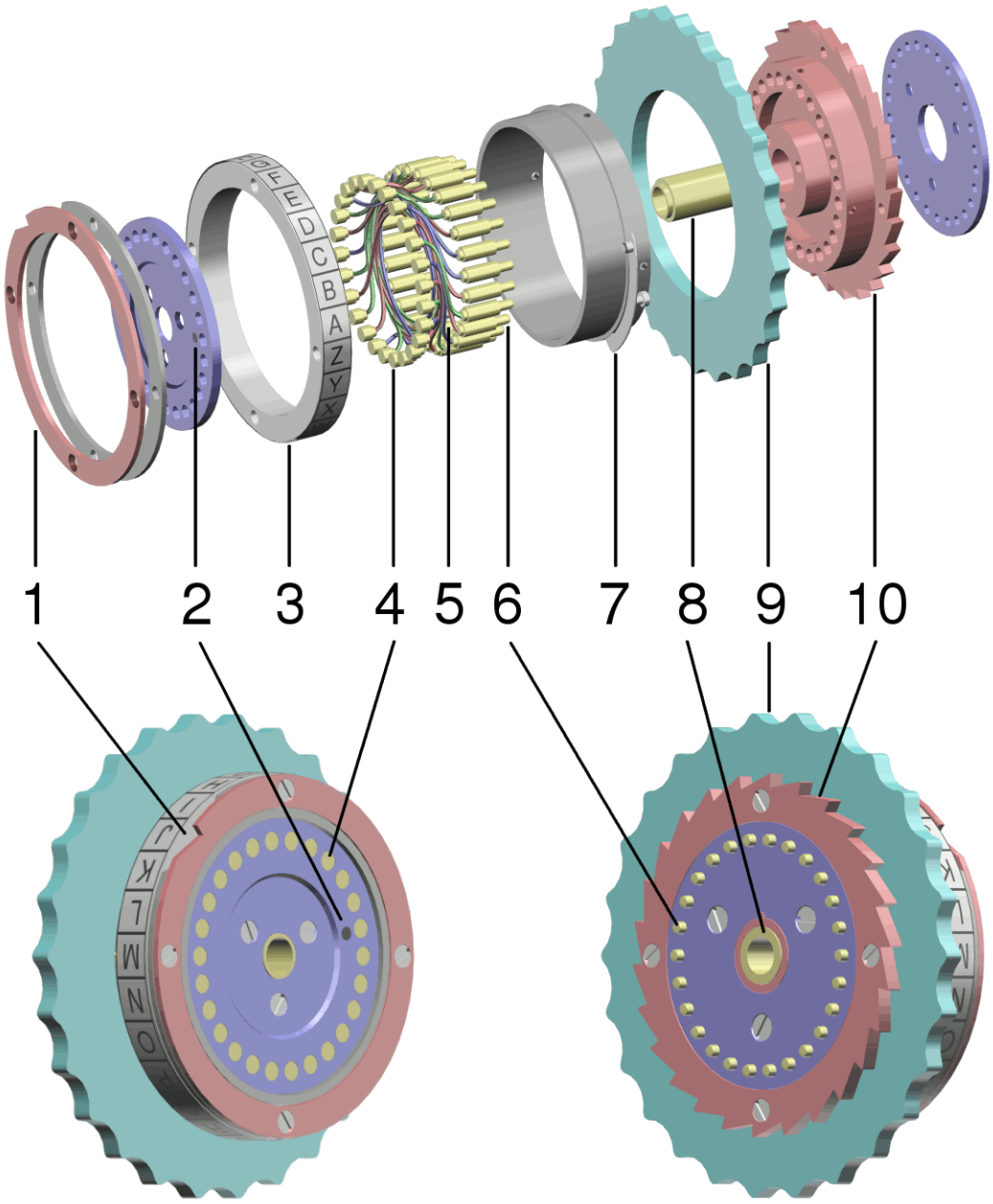






Flat contacts and notched ring

Pin contacts and ratchet wheel

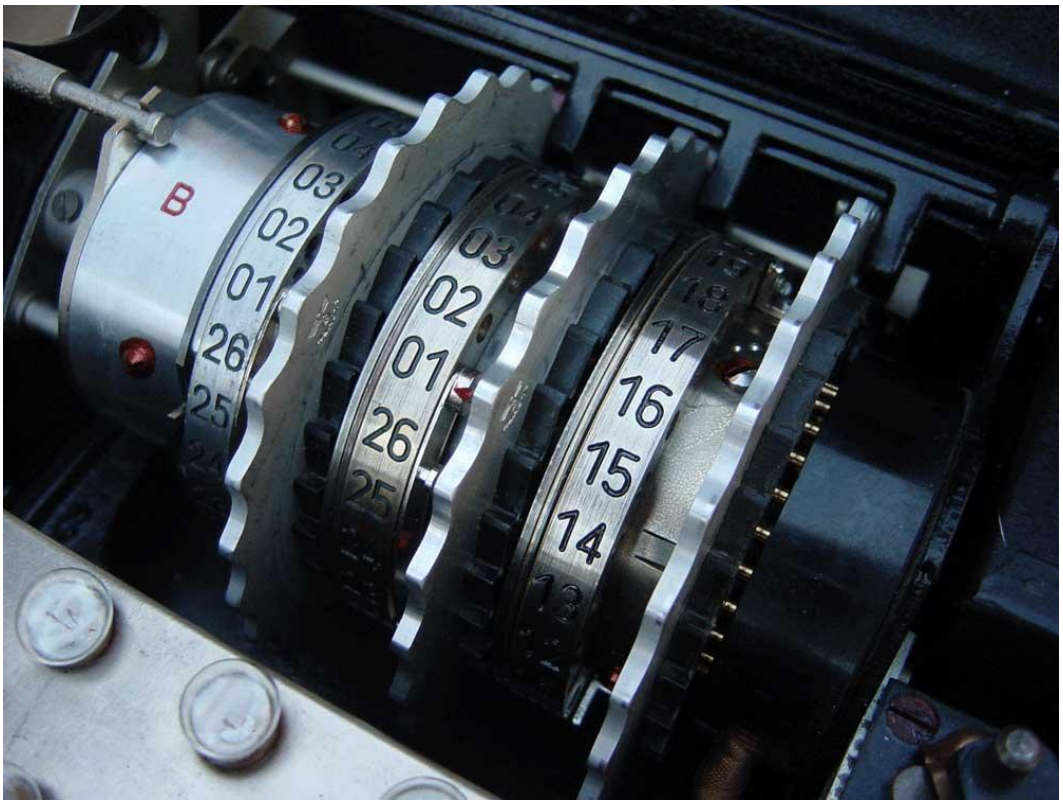


Exploded View of an Enigma's Rotor

1. notched ring
2. marking dot for "A" contact
3. alphabet ring
4. plate contacts
5. wire connections
6. pin contacts
7. spring-loaded ring adjusting lever
8. hub
9. finger wheel
10. ratchet wheel

In addition to the three rotors, there is a piece called a reflector located to the left of the rotors. This feature was distinctive of the Enigma, compared to all the other rotor encryption machines of the time. When current was passed to it from the last rotor, it redirected right back through the three rotors again. Conveniently, this made encryption the same as decryption; also called self-reciprocal ("Enigma", 2007, par 22). Like the plug board, all substitutions are one-to-one. This, unfortunately, meant that no letter could ever encrypt to itself. As a severe conceptual flaw, this made breaking the code easier.

Reflector	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	Y	R	U	H	Q	S	L	D	P	X	N	G	O	K	M	I	E	B	F	Z	C	W	V	J	A	T



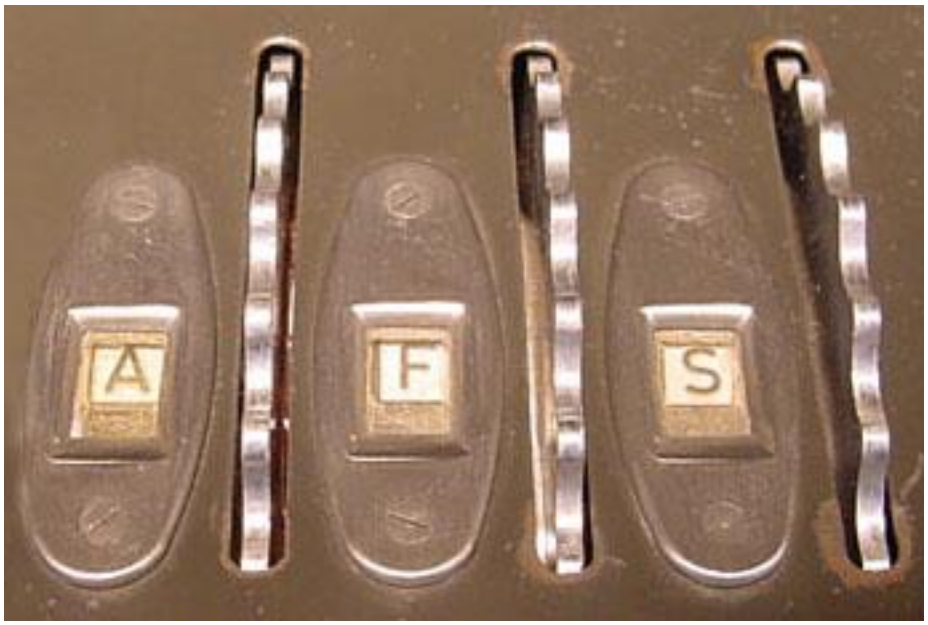
The Enigma rotor assembly. The reflector, marked B, can be seen on the far left.

After reflecting and returning through the three rotors, the signal was again affected by the configuration of the plug board. Finally, a small lamp behind the corresponding output letter was illuminated. The output indication panel, identical to the layout of the input keyboard, was located between the keyboard and the rotors on the exterior of the machine. As a more secure variant, the *Fernlesegeraet*, an accessory that attached to the Enigma in place of the indication panel, was used to prevent the operator from seeing the messages (Reuvers, 2007, pars 1-4).

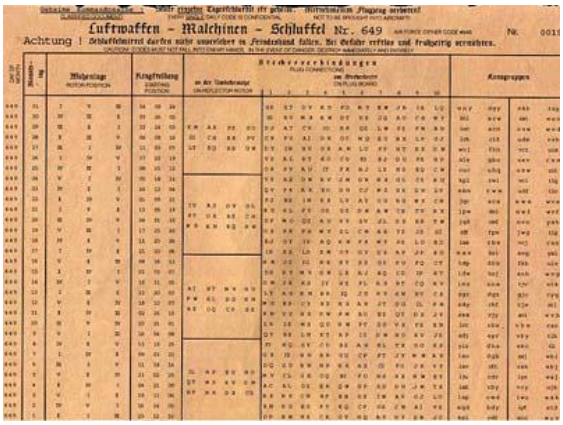


The *Fernlesegeraet*

Other accessories were also available. One was the *Schreibmax*, a small printer which transcribed the output so that the operator did not have to note which lamps were lit (“Enigma”, 2007, par 28).



Most of the settings, such as the rotor selection, reflector, and plug board, were kept constant each day. To aid in security, however, a different initial position for each for rotor was chosen specifically for each message. The starting position of each rotor was transmitted before each message, and was known as the “indicator procedure” (“Enigma”, 2007, par 35).


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The daily changes determining which rotors would be used, and how the plug board would be configured, were distributed monthly on a key sheet. This was one of the most important secrets to the operation of the Enigma, and was very closely guarded at all times. The ink used to print the codes was soluble, meaning it could be washed away with water, if its security was to be threatened (Ellis, 2005, pars 18-19).

Unknowingly, the Germans made the code easier for the British to decipher by using common greetings and salutations in their transmissions. Many messages began and ended with German phrases for “message number”, “to the group”, “nothing to report”, and “Heil Hitler!” (Ellis, 2005, pars 23-25). It has been believed that the war was shortened by two years because of the code breakers in Bletchley Park, outside of London.

The rotors and the reflector were all wired separately, which was one of the many challenges for the Allies to overcome while attempting to break the code.

Rotor	Notch	Turnover
I	Y	Q
II	M	E
III	D	V

As defined by Jim Oram (n.d.), an electronics engineer that designed a working replica of the Enigma, the *Turnover* of a rotor is “the location appearing in the window as the advance notch is engaged with the stepping lever”. The notch is then known as the actual notch on the Enigma’s rotors.

Rotor	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
I	E	K	M	F	L	G	D	Q	V	Z	N	T	O	W	Y	H	X	U	S	P	A	I	B	R	C	J
II	A	J	D	K	S	I	R	U	X	B	L	H	W	T	M	C	Q	G	Z	N	P	Y	F	V	O	E
III	B	D	F	H	J	L	C	P	R	T	X	V	Z	N	Y	E	I	W	G	A	K	M	U	S	Q	O

When each rotor steps, its list of output shifts over, similar to the logic of a Caesar shift. If rotor III starts out in the A-position ring setting in the far-right slot, it will step every key press. Therefore; theoretically, if the Enigma solely consisted of input, rotor III, and output, an input of *AAAAA* would cause an output of DFHJL. When moving from an off-set rotor to a rotor that is still on its default position, one must remember to first take the previous output and shift it in reverse.

To simplify any confusion in how the basic Enigma encryption or decryption algorithm functions, the complete process goes as follows, assuming the user was the intended recipient from which they received the transmission in Morse code:

1) The user refers to the monthly key sheet for today's setup
    a) The rotors are placed into position according to the sheet
    b) Once the rotors are in place, they are set to the day's default position
    c) The plug board is configured to swap the given characters
2) The specific message key must be decoded
    a) The rotor's rings must be set to the indicator's letters
    b) The next set of letters after the indicator's must be typed in using the indicator as the ring settings to decode the message key
3) Prepare the machine for the actual message
    a) Set the rotor's ring settings to the message key
4) Decode the actual message
    a) Type in the input, letter-by-letter, on the keyboard

b) Instantaneously with a key press, the first rotor is turned. Depending on its notch position, it may trigger the second, and even the third rotor to spin

c) A circuit is formed, and electricity from the battery is sent past the input key to the plug board

d) The current is transferred through the plug board, making switches if needed

e) The current passes through all three rotors, making three further substitutions

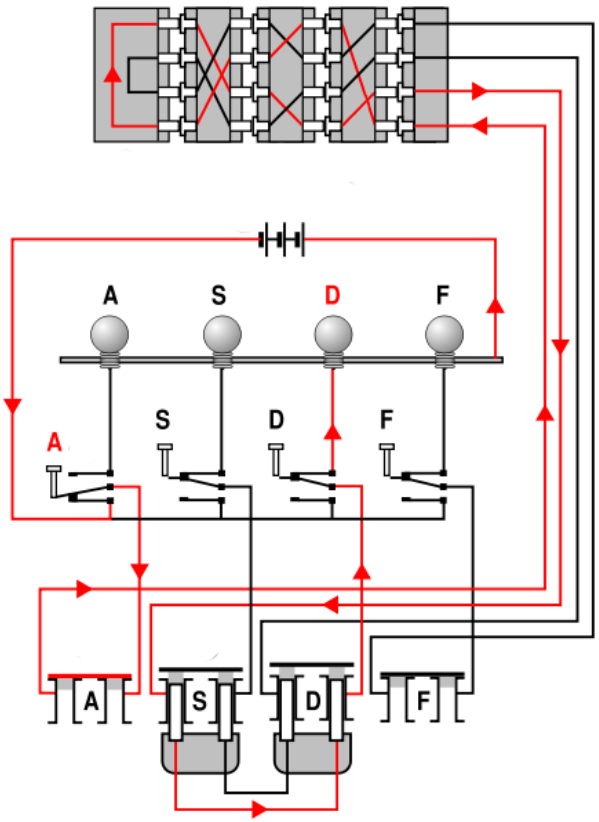
f) Via the reflector, the current is switched and sent back

g) Passing again through the three rotors, the signal is substituted three more times in *reverse*

h) If needed, another swapping at the plug board occurs

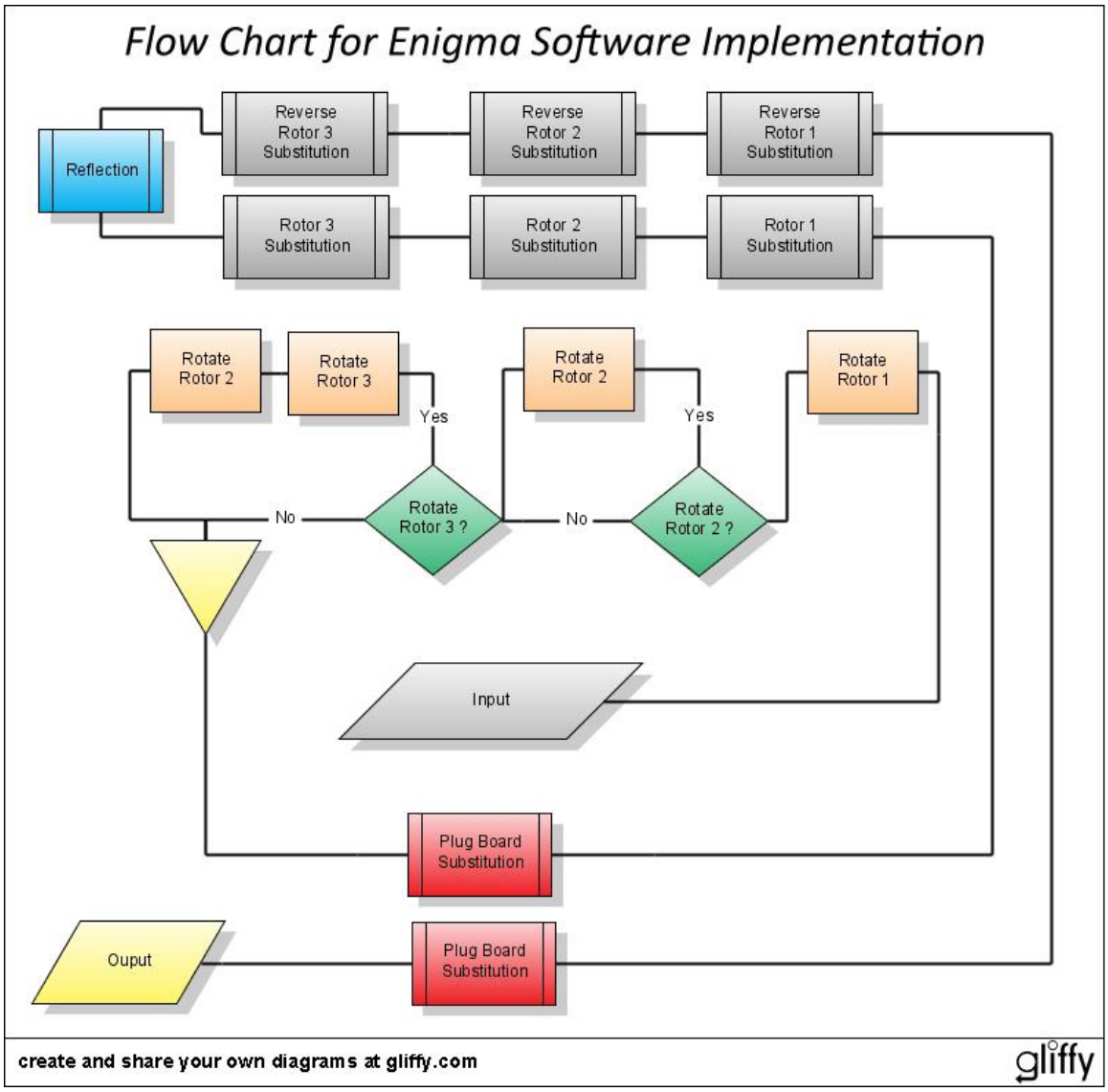
i) The light bulb completing the circuit is lit to display the output

j) This process is repeated for each and every key-stroke



Wiring diagram demonstrating current flow when the A key is pressed.

To implement the Enigma as a fully-functional software application, one would start with a Flow Chart. A flow chart is a graphical diagram to show the successive steps in a procedure. I have mapped out the logical process that would be used to create a program, based on the algorithm the Enigma uses.

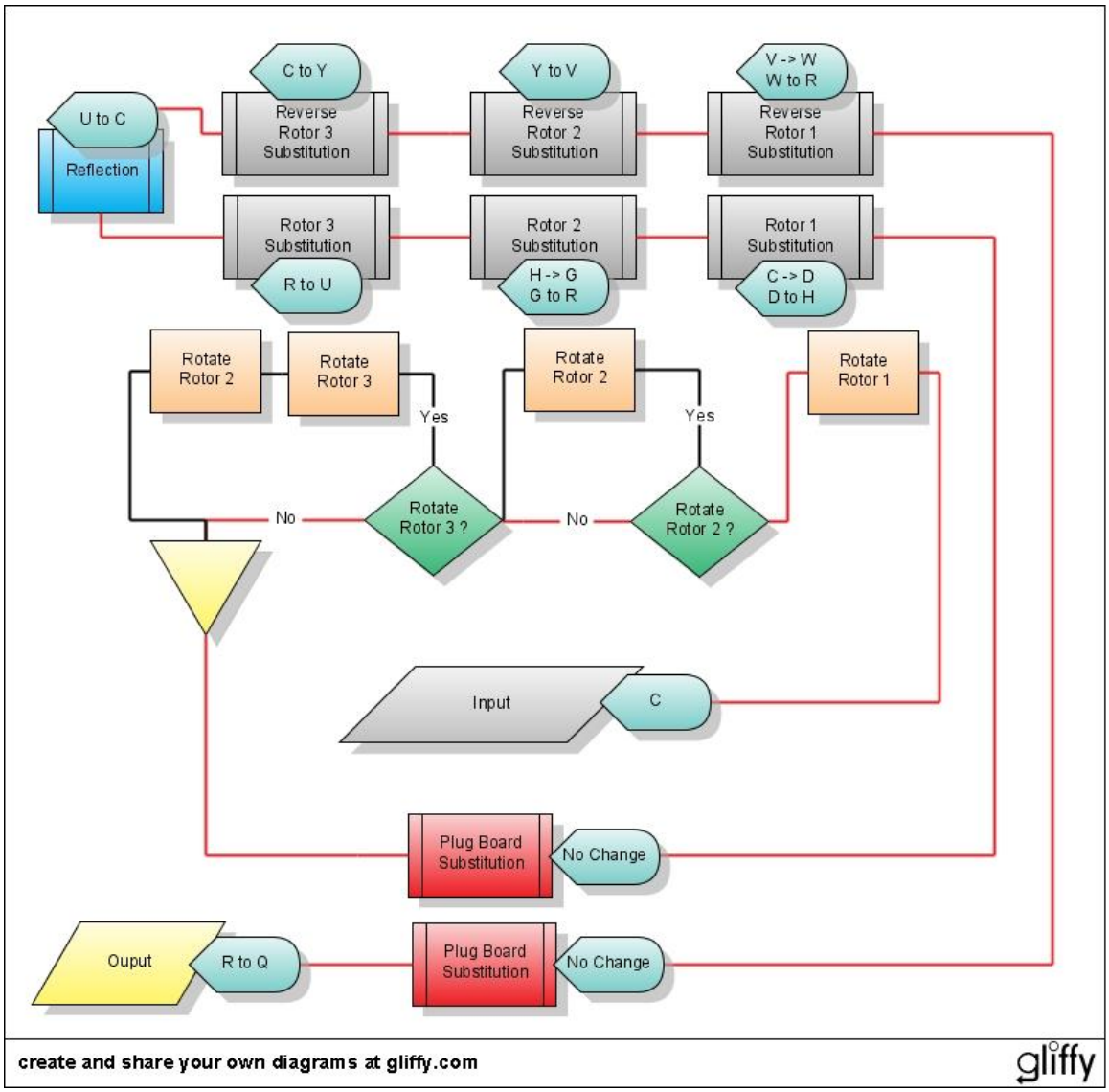


Now I will decrypt my name, CHRIS, using the Enigma encryption algorithm. We will assume that rotors I, II, and III are in order, with III being the right-most. We will also assume that they are all set so that A is showing through all of their indicator windows, and the plug board is un-altered, with A→A, B→B, C→C, etc.

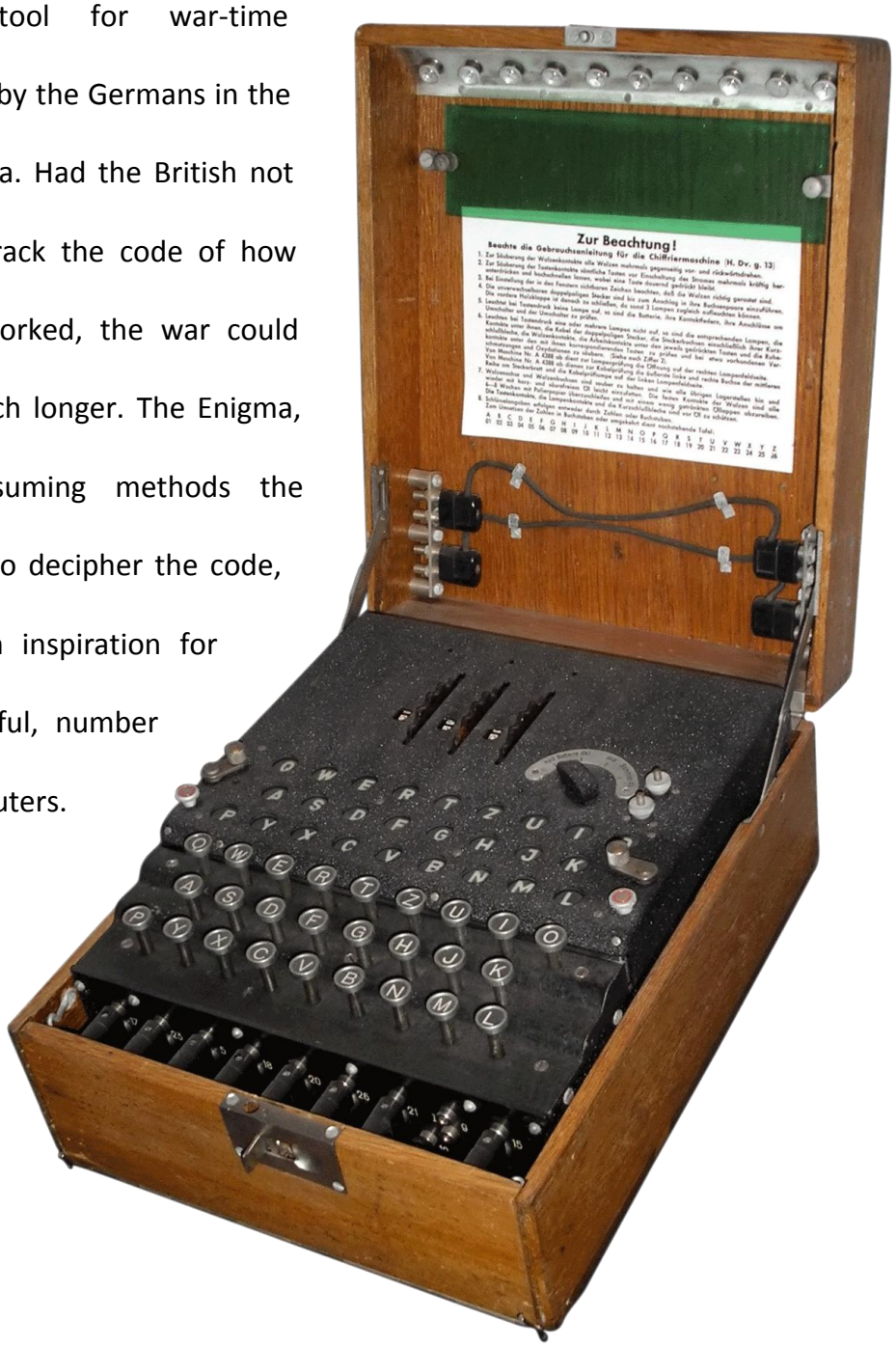
The first thing the user must do, after setting these pre-conditions, is press the "C" key. This key press, and any one after this, activates the right-most rotor to turn, immediately setting its position to B (shift of one), making the signal into a D. When D passed through rotor III, we note that it comes out H. Since rotor III is shifted and rotor II isn't, we now must convert H (via backwards shift of one) so that it is a G, and then run it through rotor II, creating an R. Rotor I is also still at its default, so the signal changes, as normally would be expected, to a U. Reflector B returns the letter C, which now must be brought back through the rotors, this time in reverse. Since we are checking in reverse, we call this an inverse function. If you were looking up the results of rotor substitution on a chart, you would start with the result, and find what it would have been if it had come through normally. In this case, C's inverse is Y, and Y's inverse is V. Now that we must shift back to the position of the off-set rotor III, we will move V to the right, using a W to calculate the inverse on rotor III, yielding an R. The output, which is never shifted, requires shifting back left, changing R into a Q. This process is repeated five times, using H, R, I, and S. The correct output for "CHRIS" using the given pre-conditions is "QPXQB". Unless the middle rotor is tripped, the process remains the same as mentioned in the

demonstration above, adding one more place to the substitution. In example, H would be shifted two places to the right, because it is now in rotor position C.

A graphical representation of this process for the letter Q encrypted to R is illustrated below using a flow chart:



Rotor encryption, specifically demonstrated by the Enigma system, is very complex. The methods and features I have described are what made this device such a powerful tool for war-time communication by the Germans in the World War II era. Had the British not been able to crack the code of how this machine worked, the war could have lasted much longer. The Enigma, the time consuming methods the British needed to decipher the code, may have been inspiration for the first powerful, number crunching computers.



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